Graph Deep Learning for Precision Glioma Mutation Mapping

Project Team

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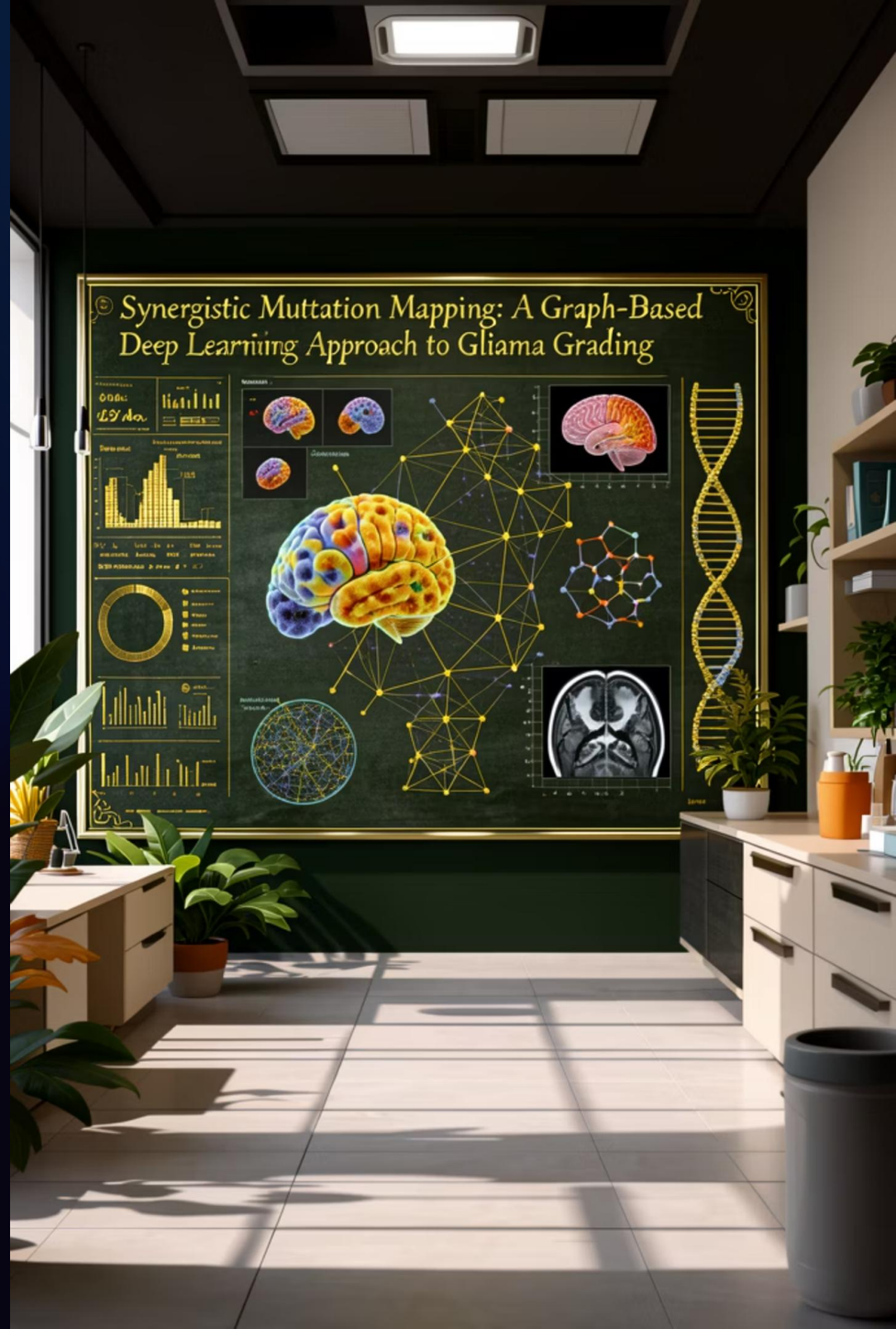


Our Approach: Synergistic Glioma Grading

1

Goal

- Detect interactions between clinical signs & mutations
- Improve glioma grade accuracy

2

Method

- Model patients as graphs of features
- Use GNNs to learn co-occurring patterns
- Predict tumor grade from learned embeddings

Dataset overview: clinical & mutation features

Source & size

UCI repository — 839 patients (LGG + GBM)

Clinical

- Age
- Sex
- Race

Genetic

- Binary calls for 20 genes (e.g., IDH1, TP53, ATRX)

Target

Predict: LGG vs GBM

Modeling mutation synergy

- **Key idea**

  Mutations often act together, not alone.

- **Goal**

  Find mutation pairs/groups that change tumor behavior.

- **Approach**

  Cross- Mutation Interaction Mapping — explicitly map interactions.

- **Why it matters**

  Reveals combined effects missed by standard classifiers.

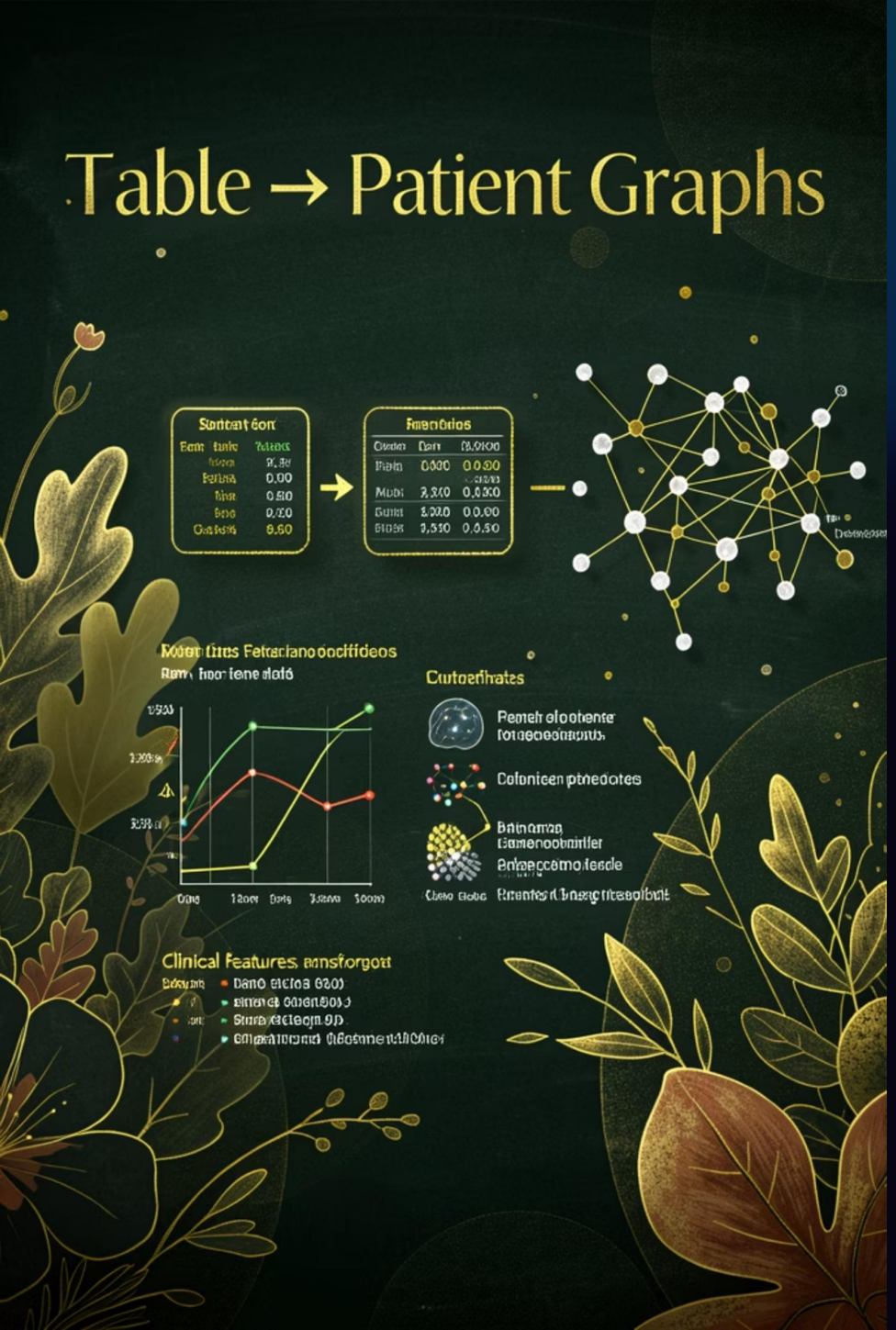


Table → Patient Graphs

Nodes (V): mutations + clinical features

Edges (E): links (co-occurrence, pathways, stats)

Patient subgraph: per- patient features & link strengths

Key takeaway: graphs preserve connections so models can use interaction patterns to predict outcomes.

Edge- Aware Message Passing (DL)

Graph convolution that uses edge weights to guide information flow.

Node updates (GCN)

- Aggregate neighbor messages to update each node.

Edge importance

- Edge weights scale message influence.

Patient embedding

- Pool node states into one vector for classification (low vs high grade).

Implementation Roadmap

01

Phase 1 — Build graph

- Find co-moving features
- Score edges with mutual information
- Incorporate expert edits

02

Phase 2 — Model comparison

- Evaluate GNN vs XGBoost & MLP
- Use cross-validation
- Track AUC, sensitivity, specificity

03

Phase 3 — Explainability

- Apply GNNExplainer
- Rank key nodes/edges
- Produce short clinician-facing notes

Evaluation Plan & Expected Outcomes

Metrics

- AUC-ROC as primary discrimination metric
- Sensitivity
- Precision & F1
- Calibration plots

Benchmarks

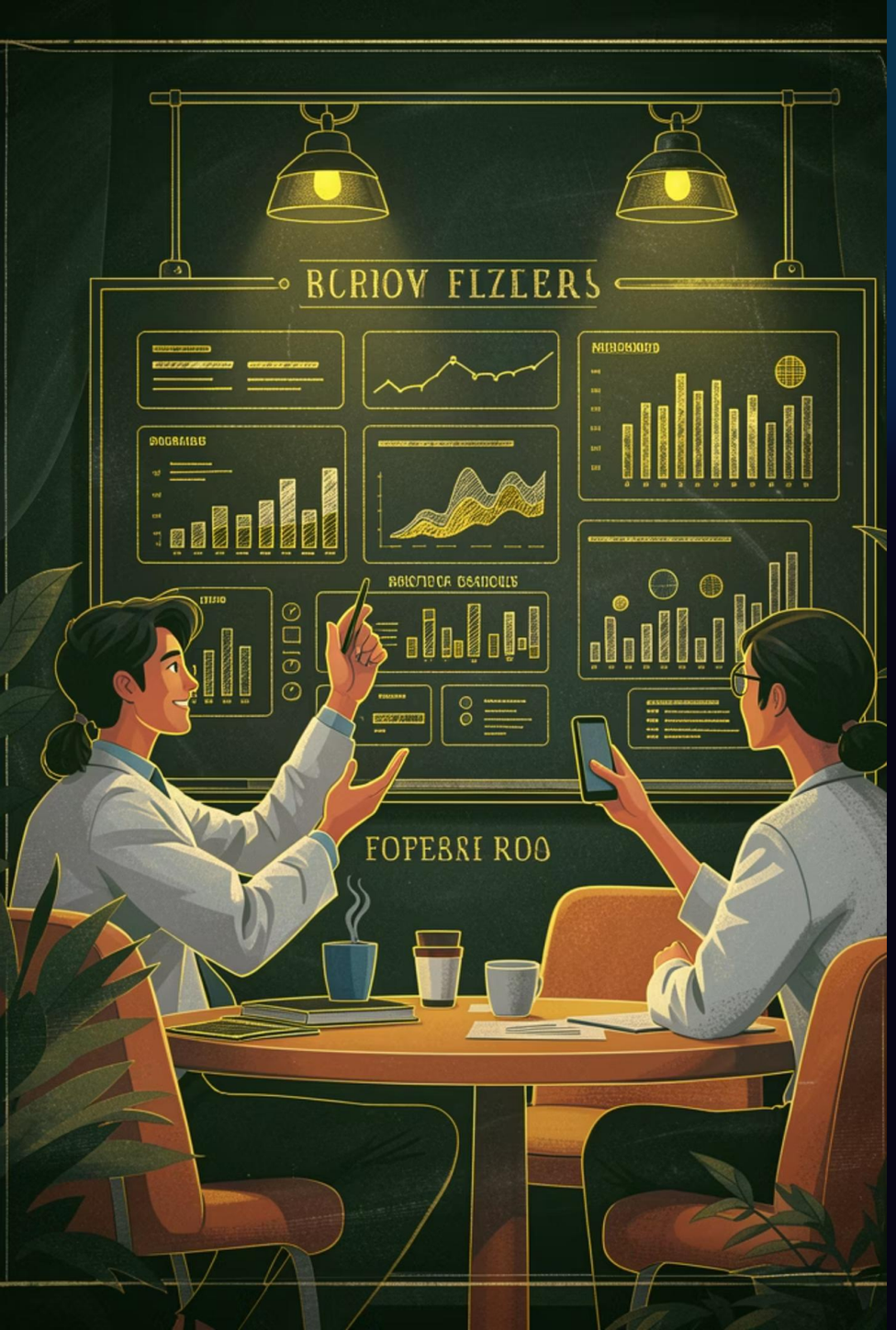
- Beat XGBoost & MLP on sensitivity/specificity
- Provide edge-level attributions (mutation pairs)

Biological validation

- Cross-check top mutation pairs with KEGG & Reactome
- Prioritize candidates for lab follow-up

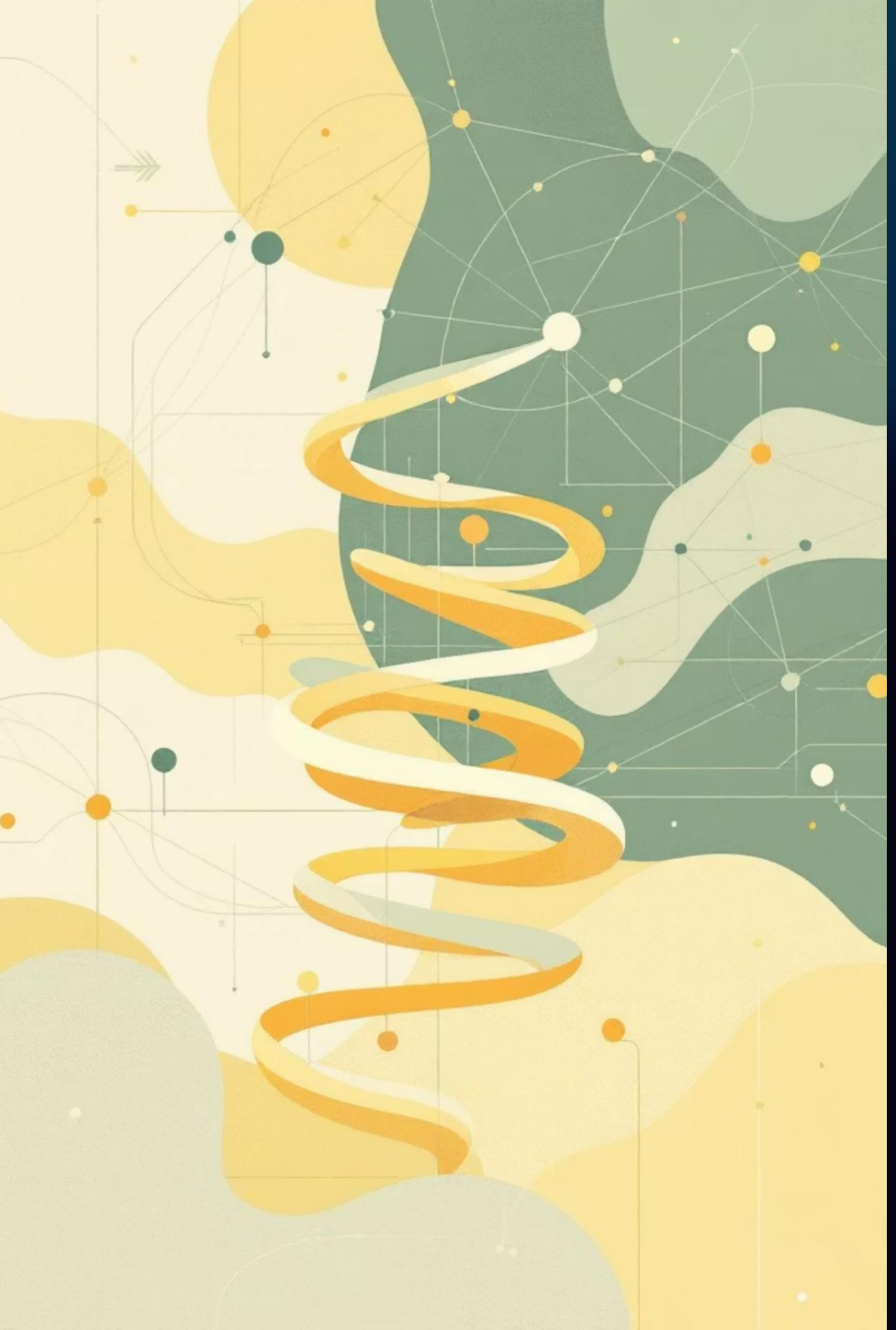
Scientific Contribution & Scalability

- *Key contributions*
  - **Models mutation–mutation interactions (not black box)**
  - **Provides edge-level explanations for testable hypotheses**
- *Scalability*
  - **Designed for large TCGA-scale cohorts**
  - **Runs on distributed clusters**



Conclusions & Next Steps

- Relational GNNs reveal gene interaction patterns missed by standard methods.
- Edge-informed message passing improves detection and gives link-level explanations.
- Next: scale to larger TCGA cohorts, run prospective validation, and pilot clinician reports.



Implementation Framework & Analytical Methods

We employ Graph Neural Networks (GNNs) to represent patients as feature graphs, using Graph Convolutional Networks (GCN) and edge-aware message passing to capture relational effects.

Model performance is benchmarked against XGBoost and multilayer perceptrons (MLPs).

To support interpretability, GNNExplainer is used to identify key mutation nodes, which are then validated against KEGG and Reactome pathways.

Dataset Overview & Research Significance

The UCI Glioma Grading Clinical and Mutation Features dataset was utilized, comprising 839 patients with a balanced distribution between LGG and GBM cases.

The dataset integrates clinical demographic variables with binary mutation indicators for 20 key genes, including IDH1 and TP53.

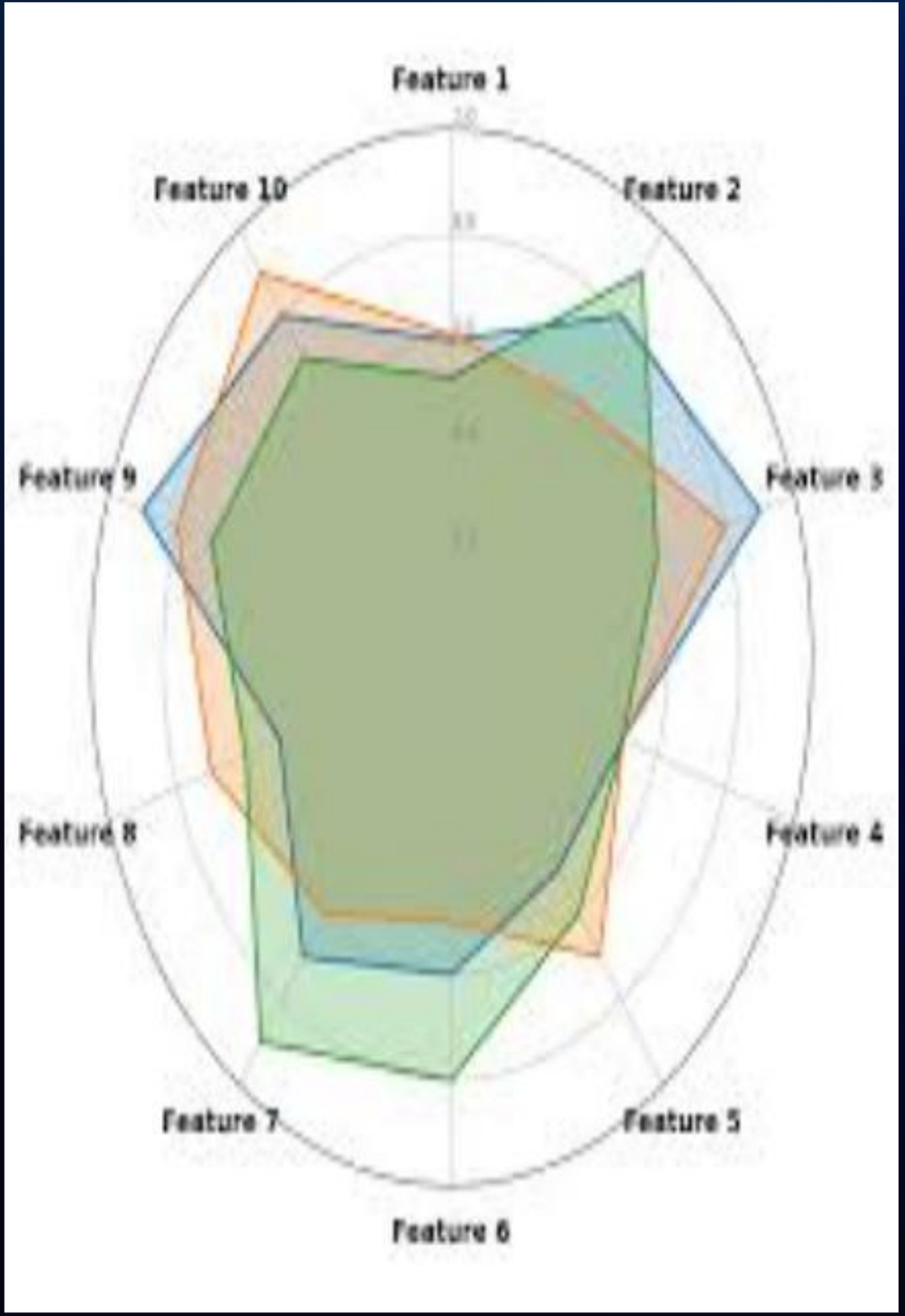
The primary objective is the development of a high-precision binary tumor-grade classification framework, with particular attention to the extent to which cross-mutation interactions may yield insights beyond those obtainable through conventional molecular testing.

Topic: Empirical Results: Latent Space Visualization (t-SNE)

- **Dimensionality Reduction: 33-D clinical/mutation features compressed into 16-D hidden states, projected to 3D space.**

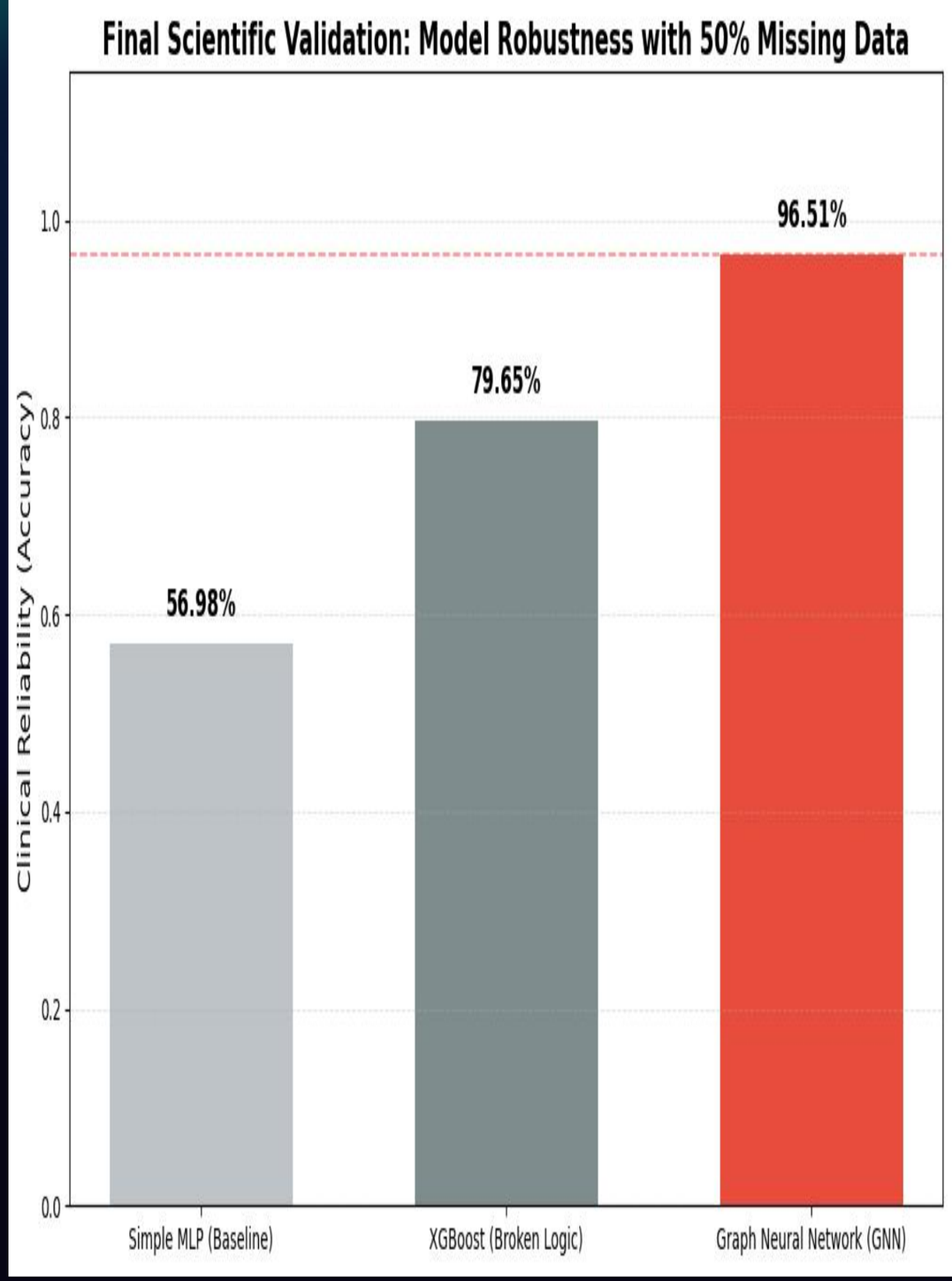
- **Class Separation: Clear spatial distinction between LGG and GBM clusters proves the GCN extracts meaningful biological structure without overfitting.**

- **Clinical Implications: Highly clustered regions indicate standard mutation profiles, while overlapping points flag atypical cases requiring detailed diagnostic review.**



Scientific Validation: Baseline Model Comparison

- **Standard MLP (Baseline): Treats patients as isolated vectors. Limited by lack of relational context.**
- **XGBoost: A powerful gradient-boosting algorithm, but fundamentally a 'shallow' learner that ignores the interconnected nature of biological data.**
- **Graph Neural Network: Uses the edge-index to aggregate features from similar patients, mirroring how oncologists look for patterns across similar clinical cases.**

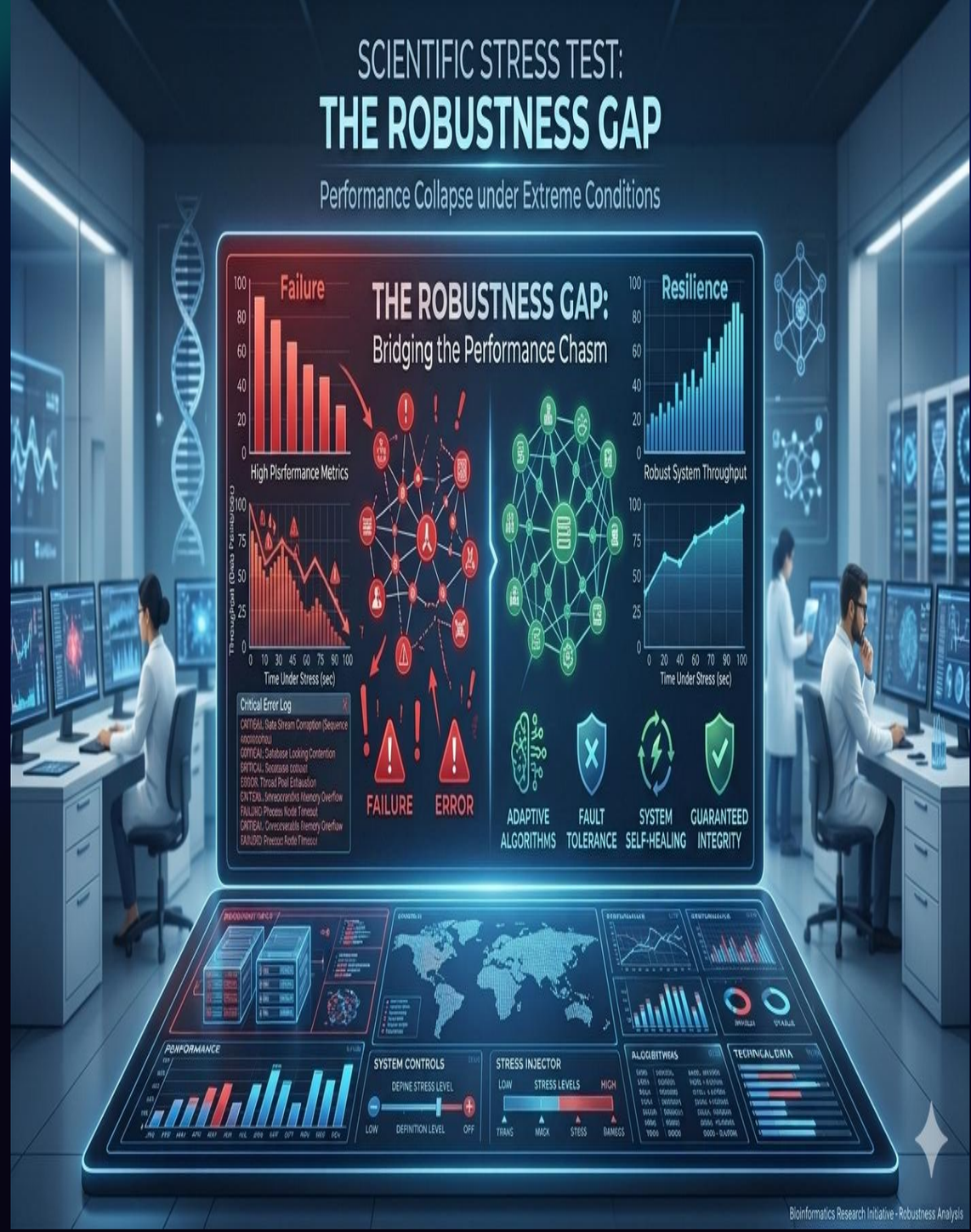


Scientific Stress Test: The Robustness Gap

- **Scenario: 50% missing data injected into the test set to simulate incomplete real-world clinical records.**

- **Results: * Simple NN (MLP): 56.98% * XGBoost (Broken Logic): 79.65% * GNN (Relational Recovery): 96.51%**

- **Scientific Conclusion: The GNN is architecturally superior. It recovers missing biological signals by aggregating data from neighboring patient nodes, making it highly reliable for clinical diagnostics.**



Thank You